Phase_3_Cross_Validation_SII_Italy_demo

October 7, 2025

1 Italian Solvency reports Table S.02.01.02; Phase 3 Cross-Validation

This script is a continuation of `Phase_2_Processing_Italy_SII_demo`.

The purpose of this script is to perform the third and final automated stage of the S.02.01.02 transcribing. Cross-Validation takes the generated table and performs a series of internal consistency checks. First on an aggregate level and then for each company in the dataset. The functions return a None if the check is passed or the fields that are compared if the check fails. This allows the user to quickly identify the mistakes in the transcribing.

The way we operate this notebook is that we first look at the summary of all companies. If the entire array is true (all tests pass for all companies), then the process moves into the final phase. If not, the user looks further down through the notebook at individual companies. Where a test fails, the relevant fields are displayed. The user than compares the pdf with the values displayed. If any corrections are necessary, the user does them above the test in code. In this way, when the script is run again, the tests passes.

1.1 Description of the process

The process of extraction is performed in 5 phases:

1.1.1 Phase 0: Find the reports and identify the relevant tables.

1) Identify the new SFCR report and save it into the folder Input.
2) Identify the pages where the tables of interest are.
3) Compile the map of the company run in the master_list.csv.

1.1.2 Phase 1: Run the Extraction script.

The script performs the following steps (with slight modifications depending on the table format): 1) Save the page with the table into a separate folder Single_pdf. 2) Use either a Python package or specialized LLM to create a digital equivalent of the table. 3) Fix the systemic errors that prevent the table from being saved as DataFrame. 4) Save the DataFrame into the Output folder.

1.1.3 Phase 2: Run the Processing script.

The script applies fixes to the DataFrame to make the numbers closer to the reported numbers. It joins all the tables into a single dataset and saves it into the Dirty_Combined folder.

1.1.4 Phase 3: Run the Cross-Validation script (this script).

Applies a series of tests that check for the internal consistency between the numbers. Flags potential errors. After the individual fixes are applied, it saves the table into the Cleaner_Combined folder.

1.1.5 Phase 4: Final modifications to the table and a manual inspection.

1.1.6 Tolerance

[1]:
```
eps = 1.5
```

Necessary packages

[2]:
```
import pandas as pd
import numpy as np
```

1.1.7 Functions

Functions that start with “check_” implement a single internal consistency check. Comparing two or more numbers if they match.

[3]:
```
def check_02_01_02_1(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0100 = R0110 + R0120

    """

    lhs = data.loc["R0100",col]
    rhs = data.loc["R0110",col] + data.loc["R0120",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        return False, pd.DataFrame(data = [data.loc["R0110",col], data.
↪loc["R0120",col], data.loc["R0100",col]], index=["R0110", "R0120","R0100"],␣
↪columns = ["TEST_1"]), diff
```

[4]:
```
def check_02_01_02_2(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0130 = R0140 + R0150 + R0160 + R0170

    """

    lhs = data.loc["R0130",col]
    rhs = data.loc["R0140",col] + data.loc["R0150",col]+ data.loc["R0160",col]+␣
↪data.loc["R0170",col]
    diff = abs(lhs-rhs)
```

```
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0140",col], data.loc["R0150",col], data.
↪loc["R0160",col], data.loc["R0170",col], data.loc["R0130",col]]
        dagnostics_index = ["R0140", "R0150", "R0160", "R0170", "R0130"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_2"]), diff
```

[5]:
```
def check_02_01_02_3(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0070 = R0080 + R0090 + R0100 + R0130 + R0180 + R0190 + R0200 + R0210

    """

    lhs = data.loc["R0070",col]
    rhs = data.loc["R0080",col] + data.loc["R0090",col]+ data.loc["R0100",col]+␣
↪data.loc["R0130",col] + data.loc["R0180",col]+ data.loc["R0190",col]+ data.
↪loc["R0200",col]+ data.loc["R0210",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0080",col], data.loc["R0090",col], data.
↪loc["R0100",col], data.loc["R0130",col],
                       data.loc["R0180",col], data.loc["R0190",col], data.
↪loc["R0200",col], data.loc["R0210",col], data.loc["R0070",col]]

        dagnostics_index = ["R0080", "R0090", "R0100", "R0130", "R0180",␣
↪"R0190", "R0200","R0210", "R0070"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_3"]), diff
```

[6]:
```
def check_02_01_02_4(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0230 = R0240 + R0250 + R0260

    """

    lhs = data.loc["R0230",col]
    rhs = data.loc["R0240",col] + data.loc["R0250",col]+ data.loc["R0260",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
```

```
    else:
        diagnostics = [data.loc["R0240",col], data.loc["R0250",col], data.
↪loc["R0260",col], data.loc["R0230",col]]
        dagnostics_index = ["R0240", "R0250", "R0260", "R0230"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_4"]), diff
```

[7]:
```
def check_02_01_02_5(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0270 = R0280 + R0310 + R0340

    """

    lhs = data.loc["R0270",col]
    rhs = data.loc["R0280",col] + data.loc["R0310",col]+ data.loc["R0340",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0280",col], data.loc["R0310",col], data.
↪loc["R0340",col], data.loc["R0270",col]]
        dagnostics_index = ["R0280", "R0310", "R0340", "R0270"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_5"]), diff
```

[8]:
```
def check_02_01_02_6(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0310 = R0320 + R0330

    """

    lhs = data.loc["R0310",col]
    rhs = data.loc["R0320",col] + data.loc["R0330",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0320",col], data.loc["R0330",col], data.
↪loc["R0310",col]]
        dagnostics_index = ["R0320", "R0330", "R0310"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_6"]), diff
```

```
[9]: def check_02_01_02_7(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
     ↪DataFrame, float]:
         """
         R0280 = R0290 + R0300

         """

         lhs = data.loc["R0280",col]
         rhs = data.loc["R0290",col] + data.loc["R0300",col]
         diff = abs(lhs-rhs)
         if diff< eps:
             return True, None, diff
         else:
             diagnostics = [data.loc["R0290",col], data.loc["R0300",col], data.
     ↪loc["R0280",col]]
             dagnostics_index = ["R0290", "R0300", "R0280"]
             return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
     ↪columns = ["TEST_7"]), diff
```

```
[10]: def check_02_01_02_8(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
      ↪DataFrame, float]:
          """
          R0500 = R0030 + R0040 + R0050 + R0060 + R0070 + R0220 + R0230 +
                  R0270 + R0350 + R0360 + R0370 + R0380 + R0390 + R0400 + R0410 +␣
      ↪R0420

          """

          lhs = data.loc["R0500",col]
          rhs = data.loc["R0030",col] + data.loc["R0040",col] + data.loc["R0050",col]␣
      ↪+ data.loc["R0060",col] + data.loc["R0070",col] + data.loc["R0220",col] +␣
      ↪data.loc["R0230",col] + data.loc["R0270",col] + data.loc["R0350",col] + data.
      ↪loc["R0360",col]+ data.loc["R0370",col]+ data.loc["R0380",col]+ data.
      ↪loc["R0390",col]+ data.loc["R0400",col]+ data.loc["R0410",col]+ data.
      ↪loc["R0420",col]
          diff = abs(lhs-rhs)
          if diff< eps:
              return True, None, diff
          else:
              diagnostics = [data.loc["R0030",col], data.loc["R0040",col], data.
      ↪loc["R0050",col], data.loc["R0060",col], data.loc["R0070",col], data.
      ↪loc["R0220",col], data.loc["R0230",col], data.loc["R0270",col], data.
      ↪loc["R0350",col], data.loc["R0360",col], data.loc["R0370",col], data.
      ↪loc["R0380",col], data.loc["R0390",col], data.loc["R0400",col], data.
      ↪loc["R0410",col], data.loc["R0420",col], data.loc["R0500",col]]
```

```
        dagnostics_index = ["R0030", "R0040", "R0050", "R0060", "R0070",␣
↪"R0220", "R0230", "R0270", "R0350", "R0360", "R0370", "R0380", "R0390",␣
↪"R0400", "R0410", "R0420", "R0500"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_8"]), diff
```

[11]:
```
def check_02_01_02_9(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0510 = R0520 + R0560

    """

    lhs = data.loc["R0510",col]
    rhs = data.loc["R0520",col] + data.loc["R0560",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0520",col], data.loc["R0560",col], data.
↪loc["R0510",col]]
        dagnostics_index = ["R0520", "R0560", "R0510"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_9"]), diff
```

[12]:
```
def check_02_01_02_10(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0520 = R0530 + R0540 + R0550

    """

    lhs = data.loc["R0520",col]
    rhs = data.loc["R0530",col] + data.loc["R0540",col]+ data.loc["R0550",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0530",col], data.loc["R0540",col], data.
↪loc["R0550",col], data.loc["R0520",col]]
        dagnostics_index = ["R0530", "R0540", "R0550", "R0520"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_10"]), diff
```

[13]:
```
def check_02_01_02_11(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
```

```
    R0560 = R0570 + R0580 + R0590

    """

    lhs = data.loc["R0560",col]
    rhs = data.loc["R0570",col] + data.loc["R0580",col]+ data.loc["R0590",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0570",col], data.loc["R0580",col], data.
↪loc["R0590",col], data.loc["R0560",col]]
        dagnostics_index = ["R0570", "R0580", "R0590", "R0560"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_11"]), diff
```

[14]:
```
def check_02_01_02_12(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0600 = R0610 + R0650

    """

    lhs = data.loc["R0600",col]
    rhs = data.loc["R0610",col] + data.loc["R0650",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0610",col], data.loc["R0650",col], data.
↪loc["R0600",col]]
        dagnostics_index = ["R0610", "R0650", "R0600"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_12"]), diff
```

[15]:
```
def check_02_01_02_13(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0610 = R0620 + R0630 + R0640

    """

    lhs = data.loc["R0610",col]
    rhs = data.loc["R0620",col] + data.loc["R0630",col]+ data.loc["R0640",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
```

```
    else:
        diagnostics = [data.loc["R0620",col], data.loc["R0630",col], data.
↪loc["R0640",col], data.loc["R0610",col]]
        dagnostics_index = ["R0620", "R0630", "R0640", "R0610"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_13"]), diff
```

[16]:
```
def check_02_01_02_14(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0650 = R0660 + R0670 + R0680

    """

    lhs = data.loc["R0650",col]
    rhs = data.loc["R0660",col] + data.loc["R0670",col]+ data.loc["R0680",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0660",col], data.loc["R0670",col], data.
↪loc["R0680",col], data.loc["R0650",col]]
        dagnostics_index = ["R0660", "R0670", "R0680", "R0650"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_14"]), diff
```

[17]:
```
def check_02_01_02_15(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0690 = R0700 + R0710 + R0720

    """

    lhs = data.loc["R0690",col]
    rhs = data.loc["R0700",col] + data.loc["R0710",col]+ data.loc["R0720",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0700",col], data.loc["R0710",col], data.
↪loc["R0720",col], data.loc["R0690",col]]
        dagnostics_index = ["R0700", "R0710", "R0720", "R0690"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_15"]), diff
```

[18]:
```python
def check_02_01_02_16(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R0850 = R0860 + R0870

    """

    lhs = data.loc["R0850",col]
    rhs = data.loc["R0860",col] + data.loc["R0870",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0860",col], data.loc["R0870",col], data.
↪loc["R0850",col]]
        dagnostics_index = ["R0860", "R0870", "R0850"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_16"]), diff
```

[19]:
```python
def check_02_01_02_17(data:pd.DataFrame, eps:float, col:str) -> list[bool, pd.
↪DataFrame, float]:
    """
    R1000 = R0500 - R0900

    """

    lhs = data.loc["R1000",col]
    rhs = data.loc["R0500",col] - data.loc["R0900",col]
    diff = abs(lhs-rhs)
    if diff< eps:
        return True, None, diff
    else:
        diagnostics = [data.loc["R0500",col], data.loc["R0900",col], data.
↪loc["R1000",col]]
        dagnostics_index = ["R0500", "R0900", "R1000"]
        return False, pd.DataFrame(data = diagnostics, index=dagnostics_index,␣
↪columns = ["TEST_17"]), diff
```

[20]:
```python
def run_check_diagnostics(table: pd.DataFrame, function, col_name: str, eps:␣
↪float) -> pd.DataFrame:
    """
    Runs a single check provided as a function for every column of a dataframe.

    """
    results = pd.DataFrame(data=[], columns=["COMPANY_NAME",col_name])
    for col in table.columns:
        res, diag, observed_tol = function(table, eps=eps, col=col)
```

```
        result_tmp = pd.DataFrame(data=[[col, res]], columns =
 ↪["COMPANY_NAME",col_name])
        results = pd.concat([results, result_tmp])
        results[results == True] =""

    return results
```

1.2 Dirty table

Upolads to memory the dirty data table from the previous phase.

[21]:
```
dirty_table_ita =  pd.read_csv("Dirty_Combined/Table_ita_S02.csv", header=0,
 ↪index_col=0)
```

[22]:
```
dirty_table_ita = dirty_table_ita.loc[dirty_table_ita.index.notna()]
```

[23]:
```
dirty_table_ita.drop(index=["CODE"], inplace=True)
```

[24]:
```
dirty_table_ita.drop(columns=["MASTER"], inplace=True)
```

[25]:
```
dirty_table_ita = dirty_table_ita.astype(float).fillna(0)
```

1.3 High-level overview of quality

Performs the checks for all companies at the same time. Assessing the overall quality of the table before looking at individual companies.

[26]:
```
results_1 = run_check_diagnostics(dirty_table_ita, check_02_01_02_1, "TEST_1",
 ↪eps)
results_1 = results_1.set_index("COMPANY_NAME")
```

[27]:
```
results_2 = run_check_diagnostics(dirty_table_ita, check_02_01_02_2, "TEST_2",
 ↪eps)
results_2 = results_2.set_index("COMPANY_NAME")
```

[28]:
```
results_3 = run_check_diagnostics(dirty_table_ita, check_02_01_02_3, "TEST_3",
 ↪eps)
results_3 = results_3.set_index("COMPANY_NAME")
```

[29]:
```
results_4 = run_check_diagnostics(dirty_table_ita, check_02_01_02_4, "TEST_4",
 ↪eps)
results_4 = results_4.set_index("COMPANY_NAME")
```

[30]:
```
results_5 = run_check_diagnostics(dirty_table_ita, check_02_01_02_5, "TEST_5",
 ↪eps)
results_5 = results_5.set_index("COMPANY_NAME")
```

[31]:
```
results_6 = run_check_diagnostics(dirty_table_ita, check_02_01_02_6, "TEST_6",␣
↪eps)
results_6 = results_6.set_index("COMPANY_NAME")
```

[32]:
```
results_7 = run_check_diagnostics(dirty_table_ita, check_02_01_02_7, "TEST_7",␣
↪eps)
results_7 = results_7.set_index("COMPANY_NAME")
```

[33]:
```
results_8 = run_check_diagnostics(dirty_table_ita, check_02_01_02_8, "TEST_8",␣
↪eps)
results_8 = results_8.set_index("COMPANY_NAME")
```

[34]:
```
results_9 = run_check_diagnostics(dirty_table_ita, check_02_01_02_9, "TEST_9",␣
↪eps)
results_9 = results_9.set_index("COMPANY_NAME")
```

[35]:
```
results_10 = run_check_diagnostics(dirty_table_ita, check_02_01_02_10,␣
↪"TEST_10", eps)
results_10 = results_10.set_index("COMPANY_NAME")
```

[36]:
```
results_11 = run_check_diagnostics(dirty_table_ita, check_02_01_02_11,␣
↪"TEST_11", eps)
results_11 = results_11.set_index("COMPANY_NAME")
```

[37]:
```
results_12 = run_check_diagnostics(dirty_table_ita, check_02_01_02_12,␣
↪"TEST_12", eps)
results_12 = results_12.set_index("COMPANY_NAME")
```

[38]:
```
results_13 = run_check_diagnostics(dirty_table_ita, check_02_01_02_13,␣
↪"TEST_13", eps)
results_13 = results_13.set_index("COMPANY_NAME")
```

[39]:
```
results_14 = run_check_diagnostics(dirty_table_ita, check_02_01_02_14,␣
↪"TEST_14", eps)
results_14 = results_14.set_index("COMPANY_NAME")
```

[40]:
```
results_15 = run_check_diagnostics(dirty_table_ita, check_02_01_02_15,␣
↪"TEST_15", eps)
results_15 = results_15.set_index("COMPANY_NAME")
```

[41]:
```
results_16 = run_check_diagnostics(dirty_table_ita, check_02_01_02_16,␣
↪"TEST_16", eps)
results_16 = results_16.set_index("COMPANY_NAME")
```

[42]:
```
results_17 = run_check_diagnostics(dirty_table_ita, check_02_01_02_17,␣
↪"TEST_17", eps)
results_17 = results_17.set_index("COMPANY_NAME")
```

[43]:
```
overall_summary = pd.
 ↪concat([results_1,results_2,results_3,results_4,results_5,results_6,results_7,results_8,res
                           ␣
 ↪results_12,results_13,results_14,results_15,results_16,results_17], axis=1)
```

[44]:
```
display(overall_summary)
```

```
                  TEST_1 TEST_2 TEST_3 TEST_4 TEST_5 TEST_6 TEST_7 TEST_8  \
COMPANY_NAME
CREDEM_VITA                                                         False
AXA
CREDIT_AGRICOLE
REALE_MUTUA
CARDIF
MEDIOLANUM                                     False
GENERALI_ITALIA    False         False         False                False
BMP_VITA
HDI                       False  False
POSTE_VITA                                                          False
INTESA_VITA
CNP_VITA                                                            False
ITAS_VITA
HELVATIA_VITA
VITTORIA
GROUPAMA                         False         False  False  False  False
ALLIANZ_UNICREDIT
ZURICH_LIFE

                  TEST_9 TEST_10 TEST_11 TEST_12 TEST_13 TEST_14 TEST_15  \
COMPANY_NAME
CREDEM_VITA
AXA
CREDIT_AGRICOLE
REALE_MUTUA
CARDIF
MEDIOLANUM                                         False
GENERALI_ITALIA
BMP_VITA           False           False
HDI
POSTE_VITA         False   False   False   False
INTESA_VITA
CNP_VITA
ITAS_VITA
HELVATIA_VITA
VITTORIA
GROUPAMA                   False           False           False
ALLIANZ_UNICREDIT
ZURICH_LIFE
```

```
                    TEST_16 TEST_17
COMPANY_NAME
CREDEM_VITA
AXA
CREDIT_AGRICOLE
REALE_MUTUA
CARDIF
MEDIOLANUM
GENERALI_ITALIA
BMP_VITA              False
HDI
POSTE_VITA
INTESA_VITA
CNP_VITA                      False
ITAS_VITA
HELVATIA_VITA
VITTORIA
GROUPAMA                      False
ALLIANZ_UNICREDIT
ZURICH_LIFE                   False
```

1.4 Manual overwrites

If a check is failing, modify the numbers at the begining of the section until the tests are passed.

Note that sometimes small errors are due to rounding.

1.5 Credemvita S.p.A.

[45]:
```python
unique_code = "CREDEM_VITA"
```

[46]:
```python
dirty_table_ita.loc["R0410",unique_code] = 82134
```

[47]:
```python
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[48]:
```python
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[49]:
```python
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[50]:
```python
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[51]:
```python
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[52]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[53]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[54]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[54]:

	TEST_8
R0030	0.0
R0040	614.0
R0050	0.0
R0060	768.0
R0070	5155576.0
R0220	4735704.0
R0230	0.0
R0270	8448.0
R0350	0.0
R0360	3402.0
R0370	1878.0
R0380	190609.0
R0390	0.0
R0400	0.0
R0410	82134.0
R0420	848.0
R0500	10179979.0

[55]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[56]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[57]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[58]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[59]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[60]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[61]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[62]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[63]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.6 AXA MPS Assicurazioni Vita

[64]:
```
unique_code = "AXA"
```

[65]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[66]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[67]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[68]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[69]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[70]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[71]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[72]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[73]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[74]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[75]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[76]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[77]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[78]:
```python
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[79]:
```python
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[80]:
```python
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[81]:
```python
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.7 CRÈDIT AGRICOLE VITA

[82]:
```python
unique_code = "CREDIT_AGRICOLE"
```

[83]:
```python
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[84]:
```python
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[85]:
```python
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[86]:
```python
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[87]:
```python
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[88]:
```python
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[89]:
```python
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[90]:
```python
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[91]:
```python
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[92]:
```python
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[93]:
```python
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[94]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[95]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[96]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[97]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[98]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[99]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.8 Società Reale Mutua di Assicurazioni

[100]:
```
unique_code = "REALE_MUTUA"
```

[101]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[102]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[103]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[104]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[105]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[106]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[107]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[108]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[109]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[110]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[111]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[112]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[113]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[114]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[115]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[116]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[117]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.9 Cardif Vita S.p.A.

[118]:
```
unique_code = "CARDIF"
```

[119]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[120]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[121]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[122]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[123]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[124]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[125]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[126]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[127]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[128]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[129]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[130]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[131]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[132]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[133]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[134]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[135]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.10 MEDIOLANUM VITA S.p.A.

[136]:
```
unique_code = "MEDIOLANUM"
```

[137]:
```
dirty_table_ita.loc["R0280", unique_code] = -1 * dirty_table_ita.loc["R0280",␣
 ↪unique_code]
dirty_table_ita.loc["R0290", unique_code] = -1 * dirty_table_ita.loc["R0290",␣
 ↪unique_code]
dirty_table_ita.loc["R0300", unique_code] = -1 * dirty_table_ita.loc["R0300",␣
 ↪unique_code]
dirty_table_ita.loc["R0630", unique_code] = -1 * dirty_table_ita.loc["R0630",␣
 ↪unique_code]
```

[138]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[139]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[140]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[141]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[142]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[143]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[144]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[145]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[146]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[147]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[148]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[149]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[150]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[151]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[152]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[153]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[154]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.11 Generali Italia S.p.A.

[155]:
```
unique_code = "GENERALI_ITALIA"
```

[156]:
```
dirty_table_ita.loc["R0100", unique_code] = 4034485
dirty_table_ita.loc["R0100", unique_code] = 4034485
dirty_table_ita.loc["R0270", unique_code] = 3446430
```

[157]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[158]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[159]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[160]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[161]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[162]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[163]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[164]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[165]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[166]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[167]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[168]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[169]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[170]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[171]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[172]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[173]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.12 Banco BPM Vita S.p.A.

[174]:
```
unique_code = "BMP_VITA"
```

[175]:
```
dirty_table_ita.loc["R0510",unique_code] = 1410
dirty_table_ita.loc["R0590",unique_code] = 7
dirty_table_ita.loc["R0860",unique_code] = 0
```

[176]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[177]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[178]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[179]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[180]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[181]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[182]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[183]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[184]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[185]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[186]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[187]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[188]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[189]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[190]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[191]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[192]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.13 HDI ASSICURAZIONI S.p.A.

[193]:
```
unique_code = "HDI"
```

[194]:
```
dirty_table_ita.loc["R0130", unique_code] = 5016257
```

[195]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[196]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[197]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[198]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[199]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[200]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[201]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[202]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[203]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[204]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[205]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[206]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[207]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[208]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[209]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[210]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[211]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.14 Gruppo Assicurativo Poste Vita

[212]:
```
unique_code = "POSTE_VITA"
```

[213]:
```
dirty_table_ita.loc["R0520", unique_code] = 236789
dirty_table_ita.loc["R0560", unique_code] = 383452
dirty_table_ita.loc["R0510", unique_code] = 620241
dirty_table_ita.loc["R0530", unique_code] = 0
dirty_table_ita.loc["R0540", unique_code] = 228156
dirty_table_ita.loc["R0550", unique_code] = 8633
dirty_table_ita.loc["R0570", unique_code] = 0
dirty_table_ita.loc["R0580", unique_code] = 361817
dirty_table_ita.loc["R0590", unique_code] = 21635
dirty_table_ita.loc["R0600", unique_code] = 135431267
```

[214]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[215]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[216]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[217]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[218]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[219]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[220]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[221]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[221]:

	TEST_8
R0030	0.0
R0040	1659986.0
R0050	0.0
R0060	25718.0
R0070	145553398.0
R0220	16973297.0
R0230	320.0
R0270	280157.0
R0350	1.0
R0360	227971.0
R0370	6468.0
R0380	8346.0
R0390	0.0
R0400	0.0
R0410	4690070.0
R0420	2355396.0
R0500	171781131.0

[222]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[223]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[224]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[225]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[226]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[227]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[228]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[229]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[230]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.15 FIDEURAM VITA S.P.A.

[231]:
```
unique_code = "INTESA_VITA"
```

[232]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[233]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[234]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[235]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[236]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[237]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[238]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[239]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[240]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[241]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[242]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[243]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[244]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[245]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[246]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[247]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[248]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

1.16 CNP Vita Assicura S.p.A.

[249]:
```
unique_code = "CNP_VITA"
```

[250]:
```
dirty_table_ita.loc["R0900", unique_code] = 23999137
```

[251]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[252]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[253]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[254]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[255]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[256]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[257]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[258]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[258]:

	TEST_8
R0030	0.0
R0040	0.0
R0050	0.0
R0060	3242.0
R0070	18123968.0
R0220	6779537.0
R0230	757.0
R0270	31280.0
R0350	0.0
R0360	6516.0
R0370	16098.0
R0380	399567.0
R0390	0.0
R0400	0.0
R0410	340483.0
R0420	60719.0
R0500	25762169.0

[259]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[260]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[261]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[262]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[263]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[264]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[265]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[266]:
```python
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[267]:
```python
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

2 ITAS VITA

[268]:
```python
unique_code = "ITAS_VITA"
```

[269]:
```python
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[270]:
```python
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[271]:
```python
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[272]:
```python
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[273]:
```python
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[274]:
```python
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[275]:
```python
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[276]:
```python
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[277]:
```python
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[278]:
```python
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[279]:
```python
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[280]:
```python
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[281]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[282]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[283]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[284]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[285]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

2.1 Helvetia Vita S.p.A.

[286]:
```
unique_code = "HELVATIA_VITA"
```

[287]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[288]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[289]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[290]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[291]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[292]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[293]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[294]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[295]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[296]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[297]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[298]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[299]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[300]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[301]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[302]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[303]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

2.2 Vittoria Assicurazioni S.p.A.

[304]:
```
unique_code = "VITTORIA"
```

[305]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[306]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[307]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[308]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[309]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[310]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[311]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[312]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[313]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[314]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[315]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[316]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[317]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[318]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[319]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[320]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[321]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

2.3 GROUPAMA ASSICURAZIONI S.P.A.

[322]:
```
unique_code = "GROUPAMA"
```

[323]:
```
dirty_table_ita.loc["R0070", unique_code] = 3780211
dirty_table_ita.loc["R0310", unique_code] = 2458
dirty_table_ita.loc["R0280", unique_code] = 78988
dirty_table_ita.loc["R0380", unique_code] = 159382
dirty_table_ita.loc["R0420", unique_code] = 151684
dirty_table_ita.loc["R0550", unique_code] = 57467
dirty_table_ita.loc["R0650", unique_code] = 1749477
dirty_table_ita.loc["R0900", unique_code] = 4518565
dirty_table_ita.loc["R1000", unique_code] = 809509
```

[324]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[325]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[326]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[327]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[328]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[329]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[330]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[331]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[332]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[333]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[334]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[335]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[336]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[337]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[338]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[339]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[340]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

2.4 UniCredit Allianz Vita S.p.A.

[341]:
```
unique_code = "ALLIANZ_UNICREDIT"
```

[342]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[343]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[344]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[345]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[346]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[347]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[348]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[349]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[350]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[351]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[352]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[353]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[354]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[355]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[356]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[357]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[358]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

2.5 Zurich Investments Life S.p.A.

[359]:
```
unique_code = "ZURICH_LIFE"
```

[360]:
```
dirty_table_ita.loc["R0600", unique_code] = 4440687
dirty_table_ita.loc["R0650", unique_code] = 4440687
dirty_table_ita.loc["R0670", unique_code] = 4368246
dirty_table_ita.loc["R0680", unique_code] = 72441
dirty_table_ita.loc["R0690", unique_code] = 5577792
dirty_table_ita.loc["R0710", unique_code] = 5530993
dirty_table_ita.loc["R0720", unique_code] = 46798
dirty_table_ita.loc["R0750", unique_code] = 51247
dirty_table_ita.loc["R0760", unique_code] = 1122
dirty_table_ita.loc["R0770", unique_code] = 25231
dirty_table_ita.loc["R0780", unique_code] = 89540
dirty_table_ita.loc["R0810", unique_code] = 3744
dirty_table_ita.loc["R0820", unique_code] = 37135
dirty_table_ita.loc["R0830", unique_code] = 3101
dirty_table_ita.loc["R0840", unique_code] = 19918
dirty_table_ita.loc["R0880", unique_code] = 30708
dirty_table_ita.loc["R0900", unique_code] = 10280225
dirty_table_ita.loc["R1000", unique_code] = 696507
```

[361]:
```
result,diagnostics, diff = check_02_01_02_1(dirty_table_ita, eps, unique_code)
diagnostics
```

[362]:
```
result,diagnostics, diff = check_02_01_02_2(dirty_table_ita, eps, unique_code)
diagnostics
```

[363]:
```
result,diagnostics, diff = check_02_01_02_3(dirty_table_ita, eps, unique_code)
diagnostics
```

[364]:
```
result,diagnostics, diff = check_02_01_02_4(dirty_table_ita, eps, unique_code)
diagnostics
```

[365]:
```
result,diagnostics, diff = check_02_01_02_5(dirty_table_ita, eps, unique_code)
diagnostics
```

[366]:
```
result,diagnostics, diff = check_02_01_02_6(dirty_table_ita, eps, unique_code)
diagnostics
```

[367]:
```
result,diagnostics, diff = check_02_01_02_7(dirty_table_ita, eps, unique_code)
diagnostics
```

[368]:
```
result,diagnostics, diff = check_02_01_02_8(dirty_table_ita, eps, unique_code)
diagnostics
```

[369]:
```
result,diagnostics, diff = check_02_01_02_9(dirty_table_ita, eps, unique_code)
diagnostics
```

[370]:
```
result,diagnostics, diff = check_02_01_02_10(dirty_table_ita, eps, unique_code)
diagnostics
```

[371]:
```
result,diagnostics, diff = check_02_01_02_11(dirty_table_ita, eps, unique_code)
diagnostics
```

[372]:
```
result,diagnostics, diff = check_02_01_02_12(dirty_table_ita, eps, unique_code)
diagnostics
```

[373]:
```
result,diagnostics, diff = check_02_01_02_13(dirty_table_ita, eps, unique_code)
diagnostics
```

[374]:
```
result,diagnostics, diff = check_02_01_02_14(dirty_table_ita, eps, unique_code)
diagnostics
```

[375]:
```
result,diagnostics, diff = check_02_01_02_15(dirty_table_ita, eps, unique_code)
diagnostics
```

[376]:
```
result,diagnostics, diff = check_02_01_02_16(dirty_table_ita, eps, unique_code)
diagnostics
```

[377]:
```
result,diagnostics, diff = check_02_01_02_17(dirty_table_ita, eps, unique_code)
diagnostics
```

2.6 Master table after cross checks

Saves the final data table in the folder `Clearer_Combined`.

[378]:
```
dirty_table_ita.to_csv("Cleaner_Combined/Table_ita_S02.csv")
```